AI STUDENT RECORD MANAGER

Step-by-Step Installation & Running Guide

Beginner-friendly guide for Windows, VS Code and Google Chrome

What this application does
Manage student records, search students, calculate totals and grades automatically, and use Groq Generative AI to produce academic performance feedback and study tips.

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1. What you need before starting

- A Windows computer with an internet connection.
- Python 3 installed on your computer.
- Visual Studio Code (VS Code).
- Google Chrome or another modern web browser.
- A Groq API key for the AI features. Student record management and result calculations can still be used if the AI service is unavailable.

2. Check the project folder

Extract the project ZIP file first. Open the extracted project folder in VS Code. The folder should contain the following files and folder:

```
AI_Student_Record_Manager_Web/
■
■■■ app.py
■■■ ai_helper.py
■■■ students.json
■■■ requirements.txt
■■■ .env.example
■■■ .gitignore
■■■ README.md
■■■ templates/
■■■ index.html
```

Important: The main Python file is app.py, not main.py. The HTML page must be named index.html and placed inside the templates folder.

3. Install Python

Visit https://www.python.org/downloads/ and download a suitable Python 3 version for Windows.

Run the installer.

On the first installation screen, tick the option “Add Python to PATH”.

Complete the installation.

Open PowerShell or the VS Code terminal and check Python:

```
python --version
```

A Python version number should appear. If Windows cannot find Python, restart VS Code after installation or check that Python was added to PATH.

4. Open the project in VS Code

Open Visual Studio Code.

Select File → Open Folder.

Choose the extracted AI_Student_Record_Manager_Web folder.

Select Terminal → New Terminal.

Make sure the terminal path ends with your project folder name.

```
PS C:\Users\YourName\Downloads\AI_Student_Record_Manager_Web>
```

5. Create a virtual environment

A virtual environment keeps this project's Python packages separate from other projects. Run this command in the VS Code terminal:

```
python -m venv venv
```

Wait for the command to finish. It creates a folder named venv inside your project folder.

6. Activate the virtual environment

For Windows PowerShell, run:

```
venv\Scripts\Activate
```

When activation succeeds, (venv) appears at the beginning of the terminal line. For example:

```
(venv) PS C:\Users\YourName\Downloads\AI_Student_Record_Manager_Web>
```

If PowerShell blocks activation: You can use Command Prompt in VS Code and run venv\Scripts\activate.bat. If your institution's computer restricts script execution, ask the lab administrator rather than changing security settings without permission.

7. Install the required libraries

With the virtual environment activated, install all packages listed in requirements.txt:

```
python -m pip install -r requirements.txt
```

The project requirements include Flask, Groq and python-dotenv. Wait until installation completes.

8. Configure the Groq API key

Open https://console.groq.com/keys and sign in or create an account.

Create an API key and copy it. Treat the key like a password.

In the project folder, make a copy of .env.example and name the copy .env.

Open .env and add your own key:

```
GROQ_API_KEY=your_actual_groq_api_key
GROQ_MODEL=
```

You may leave GROQ_MODEL empty so the application attempts to select an available text model automatically. If you choose to specify a model, use a currently supported text-generation model from your Groq account.

Security: Never share your API key, include it in screenshots, or upload the .env file to GitHub. Keep .env private. The .gitignore file should exclude it.

9. Run the application

Check that the terminal still shows (venv), then run this command:

```
python app.py
```

The terminal should display a message similar to:

```
AI Student Record Manager is running!
Open this link in Chrome: http://127.0.0.1:5000
(Press Ctrl+C in this terminal to stop the app)
```

If the browser does not open automatically, open Google Chrome and type this address in the address bar:

```
http://127.0.0.1:5000
```

Keep the terminal open while using the application. Closing the terminal or pressing Ctrl+C stops the local server.

10. How to use the application

A. Add a student

Enter the Student ID, Student Name and Class / Semester.

Enter subject names separated by commas, for example: Python, Mathematics, English.

Enter the matching marks in the same order, for example: 88, 76, 92.

Click Add. The application validates the information and saves the student record.

B. View and search records

- All saved student records appear in the Student Records table.
- Enter a student's name or ID in the search field and click Search.
- Click Show All to display all records again.
- Click a row to select that student and load their details into the form.

C. Update or delete a student

- Select a student from the table before updating or deleting.
- Edit the form and click Update to save changes. The Student ID cannot be changed.
- Click Delete and confirm the action to remove the selected record.

D. View results

The application calculates total marks, average percentage and grade automatically. The grade rules are:

Grade	Average
A	85% or above
B	70% to below 85%
C	60% to below 70%
D	50% to below 60%
F	Below 50%

E. Use the AI features

Select a student from the records table.

Click Analyze Performance to request AI feedback about strong subjects, weaker subjects and improvement suggestions.

Click Generate Study Tips to request a simple study plan and practical learning tips.

Wait for the AI response to appear in the result area. This may take a few seconds and requires an internet connection and a valid Groq API key.

11. Where are student records saved?

Student information is stored locally in students.json in the project folder. The application loads this file when it starts and saves changes when records are added, updated or deleted. Do not delete this file if you want to keep your records. Make a backup copy regularly.

12. Common problems and solutions

Problem	What to do
can't open file ... main.py	Run python app.py. The project's main file is app.py.
No module named flask	Activate venv, then run python -m pip install -r requirements.txt.
The webpage/template is not found	Check that the file is named index.html and is inside the templates folder.
The browser link does not open	Keep the terminal running and manually open http://127.0.0.1:5000 in Chrome.
Address already in use	Stop another running copy of the app with Ctrl+C, then run python app.py again.
Groq API key not found	Check that .env exists in the project root, contains GROQ_API_KEY=..., and restart the app.
Model is not available	Set GROQ_MODEL to a currently supported text-generation model in your Groq account, then restart the app.
Cannot reach AI service	Check your internet connection. Student record features should still work without the AI service.
pip installation problem	Use python -m pip install -r requirements.txt while venv is activated.

13. How to stop the application

Return to the terminal where Flask is running and press:

```
Ctrl + C
```

The local web application will stop. To run it again later, open the project folder in VS Code, activate the virtual environment, and run python app.py.

14. Student practice checklist

- ■ I can open the project folder in VS Code.
- ■ I can check my Python version.
- ■ I can create and activate a virtual environment.
- ■ I can install the project requirements.
- ■ I can configure my own Groq API key safely.
- ■ I can run app.py and open the local website.
- ■ I can add, search, update and delete a student.
- ■ I can explain how total, average and grade are calculated.
- ■ I can use both AI features and explain what they do.
- ■ I know where the JSON records are saved.